

~~max~~

$$\min x_1^2 + x_2^2 + x_3^2 = f(x)$$

$$\text{st: } x_1 + x_2 + x_3 = 1$$

$$x_1 - x_2 = 0$$

start with a feasible point

$$x_0 = (0, 0, 1)$$

$$I(x_0) = \{1, 2\} \leftarrow \text{Active constraints at } x_0$$

$$|I| = q = 2$$

$$A_q = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}_{q \times n} = 2 \times 3$$

$$A_q A_q^T = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$(A_q A_q^T)^{-1} = \begin{bmatrix} 1/3 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$p_0 = I - A_q^T (A_q A_q^T)^{-1} A_q$$

$$= \begin{bmatrix} 1/6 & 1/6 & -1/3 \\ 1/6 & 1/6 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix}$$

$$d_0 = -p \nabla f(x_0)$$

$$\nabla f(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \\ 2x_3 \end{pmatrix}$$

$$= -p \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$= - \begin{bmatrix} -2/3 \\ -2/3 \\ 4/3 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 2/3 \\ -4/3 \end{bmatrix}$$

$$x_1 = x_0 + \alpha d_0 = \begin{bmatrix} 2/3 \alpha \\ 2/3 \alpha \\ 1 - \frac{4}{3} \alpha \end{bmatrix} \in S$$

$$\begin{aligned} \min_{\alpha} \phi(\alpha) &= \min_{\alpha} f(x_0 + \alpha d_0) \\ &= \min_{\alpha} \left\{ \left(\frac{2\alpha}{3} \right)^2 + \left(\frac{2\alpha}{3} \right)^2 + \left(1 - \frac{4\alpha}{3} \right)^2 \right\} \end{aligned}$$

$$\Rightarrow \alpha = 1/2$$

$$\therefore x_1 = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

$$I(x_1) = \{1, 2\}$$

$$\nabla f(x_1) = \begin{bmatrix} 2/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

$$d_1 = -p \nabla f(x_1) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

check KKT

$$\nabla f(x_1) + \lambda_1 \nabla g_1(x_1) + \lambda_2 \nabla g_2(x_1)$$

$$= \begin{pmatrix} 2/3 \\ 2/3 \\ 2/3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 0$$

$$\begin{cases} \lambda_1 = -2/3 \\ \lambda_2 = 0 \end{cases}$$

$$\begin{cases} \lambda_1 + \lambda_2 = -2/3 \\ \lambda_1 - \lambda_2 = -2/3 \\ \lambda_1 + 2/3 = 0 \end{cases}$$

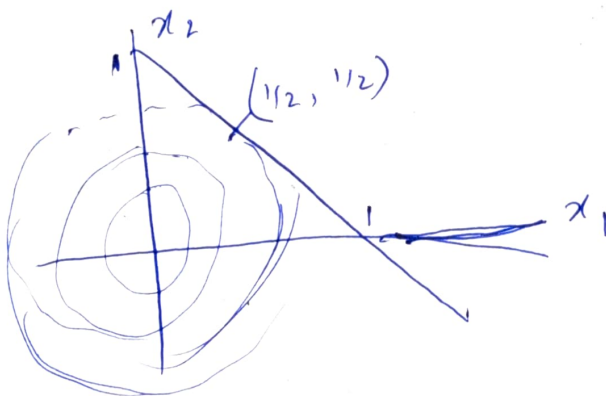
Stop

$$\min x_1^2 + x_2^2$$

ST

$$x_1 + x_2 = 1$$

$$x_1 \geq 0 \quad (-x_1 \leq 0)$$



$$\text{let } x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$I(x_0) = \{1, 2\}$$

$$\nabla f(x_0) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}$$

$$q = |I(x_0)| = 2$$

$$A_q = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}_{\substack{q \times n \\ = 2 \times 2}}$$

$$A_q A_q^T = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$P = I - A_q (A_q A_q^T)^{-1} A_q = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$d_0 = -P \nabla f(x_0) = 0$$

$$\underline{\underline{KKT}}$$

$$\nabla f(x_0) + \lambda_1 \nabla g_1(x_0) + \lambda_2 \nabla g_2(x_0) = 0$$

$$\begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \end{pmatrix} = 0$$

$$\lambda_1 - \lambda_2 = 0 \Rightarrow \lambda_1 = -2$$

$$\lambda_1 + 2 = 0 \Rightarrow \lambda_2 = -2 < 0$$

not KKT point

proceed with method:
modify A_q by removing 2nd row
from A_q

$$\hat{A}_q = [1, 1]$$

$$\hat{A}_q \hat{A}_q^T = 2$$

$$P = I - \frac{1}{2} \hat{A}_q^T \hat{A}_q = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

$$d_0 = -P \nabla f(x_0)$$

$$= -P \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore x_1 = \begin{pmatrix} \alpha \\ 1-\alpha \end{pmatrix}$$

$$[x_0 + \alpha d_0]$$

condition : $\alpha > 0$

$$\phi(\alpha) = f(x_0 + \alpha d_0)$$

$$= \alpha^2 + (1-\alpha)^2$$

$$= 2\alpha^2 - 2\alpha + 1$$

min $\alpha > 0$ $\phi(\alpha)$

$$\phi'(\alpha) = 0$$

$$\alpha = 1/2 (> 0)$$

$$\therefore x_1 = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

$\hookrightarrow A|_{x_1} = \begin{bmatrix} 1 & 1 \end{bmatrix}$

$$\Gamma(x_1) = \{1\}$$

($q=1$)

(same P because same $A\hat{q}$)

$$d_1 = -P \nabla f(x_1)$$

$$= - \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

check KKT (for active constraints)

$$\nabla f(x_1) + \lambda_1 \nabla g_1(x_1) = (1) + \lambda_1 (1) = 0$$

$$\underline{\lambda_1 = -1}$$

$\therefore$ KKT point

$\therefore$ Stop

Non Linear Constraints

$$\min f(x)$$

$$\text{ST } \begin{aligned} g_i(x) &\leq 0 & i = 1 \dots m \\ h_j(x) &= 0 & j = 1 \dots p \end{aligned}$$

g_i are convex $g_i \in C'$
 h_j are affine funcⁿ $h_j \in C'$
 f is convex $f \in C'$

penalty function approach

$$f(x) + \alpha P(x)$$

$\uparrow$
hyper parameters

$\downarrow$
unconstrained problem

$$P(x) = \begin{cases} 0 & x \in S \\ \infty & x \notin S \end{cases}$$

$\uparrow$
penalty funcⁿ

$(M \gg 0)$

this penalty funcⁿ is not smooth.

$$P(x) = \sum_{i=1}^m \max \{0, g_i(x)\}^2 + \sum_{j=1}^p \max \{0, h_j(x)\}^2 + \max \{0, -h_j(x)\}^2$$

$$\nabla P(x) = \sum 2 \max \{0, g_i(x)\} \nabla g_i(x)$$

$$+ \sum_{j=1}^p 2 (h_j(x)) \nabla h_j(x)$$

original problem is equivalent to solving
 $\min_{x \in \mathbb{R}^n} (f(x) + \alpha P(x)) = q(x, \alpha)$

↑ unconstrained
↑ differentiable

$\alpha > 0$ is
hyper parameter

take any arbitrary $x_0 \in \mathbb{R}^n$ $\alpha_0 > 0$

solve it by some iterative scheme
starting from x_0 . (α fixed)

Stop when $\| \nabla_x q(x, \alpha_0) \| < \varepsilon_0$

suppose stop @ k^{th} iteration.

we have x_k

now choose $\alpha_1 > \alpha_0$

now $\left[\min_x q(x, \alpha_1) \right]$

Solve this problem by iterative scheme

start from $x_0 = x_k$

stop when $\| \nabla_x q(x, \alpha_1) \| < \varepsilon_1$

$(\varepsilon_1 < \varepsilon_0)$ why??

Suppose we stop @ l^{th} iteration to get x .

choose $\alpha_2 > \alpha_1$

Solve $\min_x q(x, \alpha_2)$ starting from x

stop when $\| \nabla_x q(x, \alpha_2) \| < \varepsilon_2$ ($\varepsilon_2 < \varepsilon_1$)

and so on.

[Stop when penalty funcⁿ
becomes zero ($< \varepsilon$)

inner problem: α_k, ϵ_k

outer problem: } tolerance for original problem

α_k strictly $\uparrow$ +ve seq

ϵ_k strictly $\downarrow$ +ve seq